

TASK 6: Sales Trend Analysis Using Aggregations

Objective: Analyze monthly revenue and order volume.

Tools: PostgreSQL / MySQL / SQLite

Deliverables: SQL script + results table

Hints/Mini Guide:

- Use `EXTRACT(MONTH FROM order_date)` for month.
- `GROUP BY` year/month.
- Use `SUM()` for revenue.
- `COUNT(DISTINCT order_id)` for volume.
- Use `ORDER BY` for sorting.
- Limit results for specific time periods.

Dataset: online_sales (orders table with order_date, amount, product_id)

Outcome: Learn how to group data and analyze time trends.

Interview Questions:

- How do you group data by month and year?
- What's the difference between `COUNT(*)` and `COUNT(DISTINCT col)`?
- How do you calculate monthly revenue?
- What are aggregate functions in SQL?
- How to handle NULLs in aggregates?
- What's the role of `ORDER BY` and `GROUP BY`?
- How do you get the top 3 months by sales?

Task Submission Guidelines

- **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM.

- **Self-Research Allowed:**

You are **Debug** free to **Yourself**: explore, Google, or refer to tutorials to understand concepts and complete the task effectively. Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a **short README.md** explaining what you did.

Submit Here:

After completing the task, paste your GitHub repo link and submit it using the link below:

SUBMISSION LINK

